INDIA'S ANTI-SATELLITE PROGRAMME: New Opportunities For Industry

Just imagine an Indian spy satellite intercepting enemy signal that indicates imminent land, sea, air and cyber-attacks on critical defence positions and installations. The Chief of Defence Staff (CDS) holds an emergency four-way conference with ground, sea, and air commanders thousands of kilometres apart through a satellite phone. Low-orbiting surveillance satellites locate the enemy battle formations, radar installations, and missile pads. The army formations, combat aircraft, and nuclear submarines get into action using navigation and communication satellites. This was one of the scenarios played out in India's first simulated space warfare exercise, IndSpaceEx, conducted recently by the Indian armed forces. The exercise provided all stakeholders a deep understanding of the emerging space security challenges to review the existing resources and plan appropriate defence capabilities and assets in space

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Today, the military planning and execution is very much dependent on space based systems. Every aspect of warfare, from strategic targeting of intercontinental ballistic missiles to covert operations by special forces, has an essential space link. Satellites are being extensively used for communication, missile warning, weather information, imagery and intelligence, surveillance and reconnaissance, and services such as providing information related to position, navigation, and timing.

The satellites are the eyes, ears and voice of the modern military commanders. Realising that future wars will be controlled from space using electromagnetic spectrum, nations are vying to gain an edge in space. Space has become the definitive 'high ground' where tomorrow's wars may be won or lost and countries are moving towards the application of military force in, from, and through space.

Fig. 1: Depiction of Indian Army's Battlefield Management System (Source: https://spslandforces.com)

In India, all space situational awareness related tasks extending to tactical, predictive, and intelligence have been placed under The Space Situational Awareness Directorate using an integrated battle management command, control, and communications (BMC3) architecture. ISRO and DRDO, in cooperation with public and private sectors, have been entrusted with the task of developing the requisite technology and assets to address the technical demands and necessities of the armed forces.

The line between militarisation and weaponisation is quite blurred. Militarisation broadly encompasses any activity to build up to a state of conflict, whereas weaponisation, by contrast, means actively developing or deploying a weapon. The militarisation of space assists armies on the conventional battlefield, whereas weaponisation of space enables the outer space itself to become the battleground, which is sometimes referred to as the